NBA and EuroLeague basketball, 2010/11–2025/26: a Bayesian replication and update

Loukas Theodosiou
TODO: affiliation
loukesio@gmail.com

2026-11-03

Abstract Mandić and colleagues (2019) compared the NBA and the EuroLeague from 2000 to 2017 and concluded that the two leagues were becoming more alike. We re-implement their model in Stan, reproduce all 68 of their published estimates from their own data (largest difference 0.013), and extend the comparison to sixteen seasons (2010/11–2025/26), 24,874 games and 4,057,782 located shots. Bayesian mixed models with team- and opponent-season effects describe the leagues today; second-order random-walk trends, a shift model with a discrete lag and a "time gap" estimate describe how they changed; a matched design compares playoffs with the same teams' regular seasons; and shrinkage style profiles place every team and player on its league's path through time. Today the two leagues shoot equally well and take the same share of their shots from three (41.1% in both), but they differ in style: the NBA plays 10.4 more possessions per 40 minutes and the EuroLeague commits 7.5 more fouls per 100 possessions. Shot mix converged while pace diverged. On three-point shooting the EuroLeague moves with the NBA or a season behind it; on volume measures it resembles the NBA of three to five seasons ago. Houston (NBA) and Real Madrid (EuroLeague) played their leagues' future most clearly. Code, models and figures are openly available.

1 Introduction

The NBA and the EuroLeague are the two strongest basketball competitions in the world, but they are played under different rules (a 48- against a 40-minute game, a 7.24 m against a 6.75 m three-point line, different defensive and foul rules) and by teams built in different ways. How the two leagues differ, and whether they are becoming alike, matters to coaches and scouts who move players between them and to analysts who compare their statistics.

Mandić et al. (2019) compared the leagues from 2000/01 to 2016/17 with a Bayesian model of season means and linear trends. They found that the NBA was moving towards the EuroLeague's style, most clearly in three-point shooting, and that NBA playoff basketball looked more "European" than the NBA's regular season. Since 2017 the three-point revolution has continued (Goldsberry, 2019), both leagues have changed rules (the 14-second shot-clock reset after an offensive rebound reached the NBA in 2018/19), and shot-location data have become available for both leagues, which allows comparisons the original box-score study could not make (Erčulj & Štrumbelj, 2015).

This paper replicates and extends that study. We ask four questions:

- **RQ1.** How do the two leagues differ today (2023/24–2025/26), in box scores and in where shots are taken and made?
- **RQ2.** How have they changed since 2010/11? Are they converging, and does one league follow the other?
- **RQ3.** How does play change in the playoffs, and do NBA playoffs look more like the EuroLeague?
- **RQ4.** Which teams and players embodied their league's typical style, and which were ahead of their time?

The contributions are (i) a full replication of the 2019 model on the authors' own data; (ii) an update to 2025/26 with models that separate team, opponent and season effects; (iii) new analyses of convergence, lead–lag and playoff change; and (iv) a probabilistic "time position" for every team and player. All code, models and figures are open.

2 Data

Box scores come from ESPN for the NBA, through the hoopR package (Gilani, 2021), and from the official EuroLeague API for the EuroLeague. Shot locations come from NBA Stats (through hoopR) and from the EuroLeague API. The main window is 2010/11–2025/26: 24,874 games and 4,057,782 field-goal attempts with court coordinates. Seasons from 2001/02 are used only for the lead–lag model. Seasons are labelled by their starting year in the code (2010 = 2010/11).

The metrics follow Mandić et al. (2019) exactly. Possessions are 2PA + 3PA + 0.5·FTA + TOV; counting statistics are per 100 possessions; the four factors (Kubatko et al., 2007; Oliver, 2004) are computed for each team and its opponent; free throws are counted per foul received; and rotation depth is the entropy of minutes played. We add possessions per 40 minutes (to put the 48- and 40-minute games on one scale), points per 100 possessions and the share of field-goal attempts taken from three.

Two scorekeeping systems. Rebounds and turnovers are summed over players in both leagues, because team rebounds and team turnovers are recorded differently. EuroLeague scorers place layups and dunks about 1.1 m from the hoop on average, so close-range EuroLeague shots appear as a ring rather than at the rim; the rim area is therefore defined as within 1.75 m. Shots are grouped into five zones: rim area (< 1.75 m), short mid-range (1.75–4.25 m), long mid-range (two-pointers beyond 4.25 m), corner three and above-the-break three, each with its own league's court geometry.

Playoffs. NBA playoffs exclude the play-in. EuroLeague playoffs are the playoff series and the Final Four; the former Top 16 counts as regular season (the 2019 paper counted it as playoffs) and the 2025/26 play-in is dropped.

3 Methods

All models are Bayesian and fitted in Stan (Carpenter et al., 2017), directly or through brms (Bürkner, 2017). Every reported model passed the usual checks (R-hat ≤ 1.04, no or negligible divergent transitions). Figures were drawn with the ggbasket package (Theodosiou, 2026).

Replication (RQ2). The 2019 model describes each league and metric by game-level values $y_{ij} \sim N(\mu_j, \sigma_j)$ in season j, with season means on a line, $\mu_j \sim N(\mu + s\,t_j, \sigma)$, and flat priors. Because games are independent and normal within a season, the season's game count, mean and variance are sufficient statistics; we write the model on them, which gives the same posterior about a thousand times faster. Fitted to the authors' published data (PLOS ONE S1 Data), it reproduces their Table 1 (Section 4.2).

Differences today (RQ1). Each metric is standardised and modelled at team-game level as $z \sim N\big(\beta_{\text{league}} + u_{\text{team-season}} + v_{\text{opponent-season}}, \sigma_{\text{league}}\big)$, with weakly informative priors ($N(0, 1.5)$ on league effects, $N(0, 1)$ on standard deviations). The league difference is reported in game-to-game standard deviations and on the original scale. Shot shares and accuracy by zone use binomial models with team-season effects; accuracy across the court uses a binomial model with a smooth two-dimensional tensor-product surface per league (Wood, 2017).

Trends and convergence (RQ2). Season means are modelled as a second-order random walk (RW2) (Rue & Held, 2005) plus season-specific shocks, $\bar{y_t} \sim N\big(f_t, \sqrt{se_t^2 + \tau^2}\big)$ with $f_t = 2f_{t-1} - f_{t-2} + \omega\,\varepsilon_t$. When $\omega \to 0$ the trend is the straight line of the 2019 model. Because all terms are Gaussian, the curve is integrated out analytically and only ω and τ are sampled; curves are then drawn from their exact conditional distribution. The gap between the leagues' curves, and the probability that it shrank, follow from the joint draws.

Lead–lag (RQ2). A shift model asks whether the EuroLeague's trend equals the NBA's shifted by k seasons, $\bar{y_{t^E}} \sim N(a + b\,f_{t-k}, \cdot)$, with $k \in \{-5, \ldots, 5\}$ marginalised out, so the posterior $P(k \mid \text{data})$ says how

strongly each lag is supported. Probability that piles up at the edges of the window means that no lag fits well. A second, descriptive measure finds, for each posterior draw of the NBA curve, the season in which the NBA first reached the EuroLeague's 2025/26 level ("time gap").

Playoffs (RQ3). Only teams that reached the playoffs are used, each compared with its own regular season, with team- and opponent-season effects: $z \sim N\left(\beta_{\text{league}} + \gamma_{\text{league}} \cdot \text{playoff} + u + v, \sigma\right)$. For each metric we also compute the probability that the NBA's playoff shift points towards the EuroLeague's regular-season level.

Style profiles (RQ4). Each team-season is described by 14 numbers (pace, five-zone shot mix, free throws, assists, turnovers, offensive rebounds, blocks, steals, fouls and rotation depth) and each rotation player-season by 13, always compared with players of the same position. Profiles are conjugate posteriors (Dirichlet–multinomial for shot mix, gamma–Poisson for rates, normal for game means) with prior strength estimated by marginal likelihood (empirical Bayes), so small samples are shrunk towards the league–season average. A team's *time position* is the season on its league's path through style space that it is closest to; a team of 2012/13 closest to the league of 2016/17 is four seasons ahead.

4 Results

4.1 How the leagues differ today (RQ1)

The two leagues now shoot alike (Figure 1). Three-point percentage (36.1% NBA, 36.1% EuroLeague), free-throw percentage, effective field-goal percentage and the share of shots taken from three (41.1% against 41.1%) are practically identical. They differ in style. The NBA plays faster (94.2 against 83.8 possessions per 40 minutes), blocks more and earns more free throws per foul (1.16 against 0.90). The EuroLeague fouls far more (24.3 against 16.8 per 100 possessions), turns the ball over more and rebounds more of its own misses.

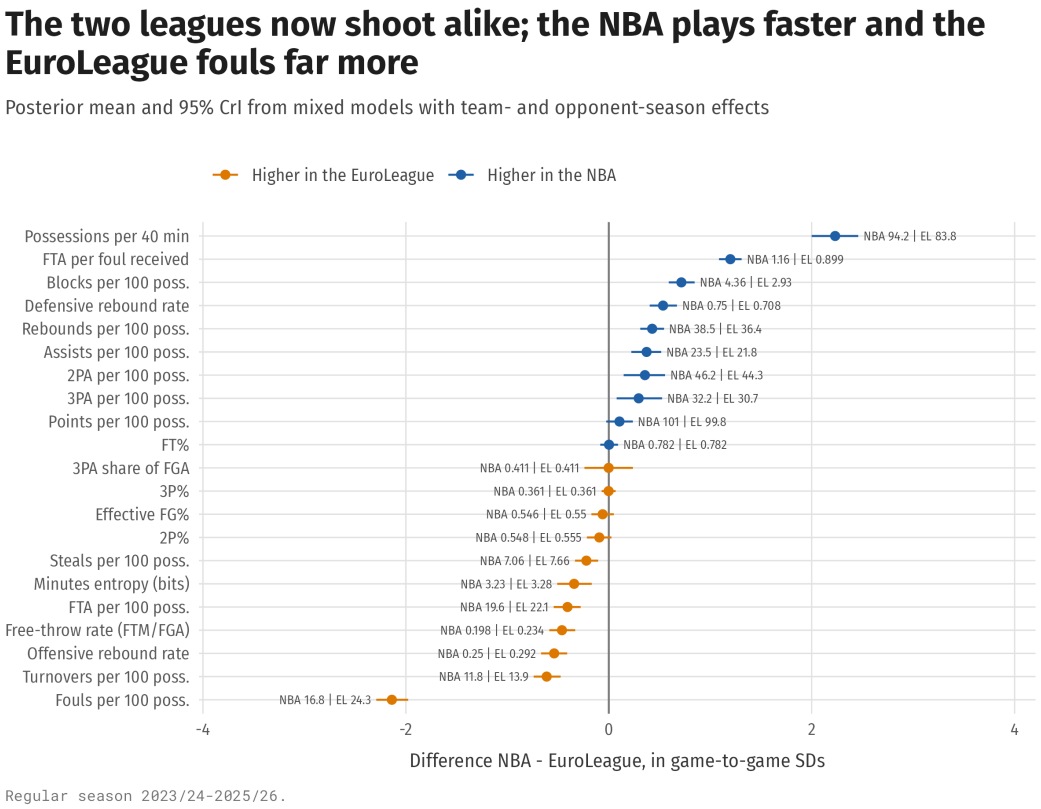


Figure 1: Box-score differences between the leagues, regular season 2023/24–2025/26. Posterior means and 95% credible intervals from mixed models with team- and opponent-season effects, in game-to-game standard deviations; numbers are the leagues' values on the original scale.

The equal three-point share hides different shot locations (Figure 2, Figure 3). 11% of NBA shots come from the corner against 4% in the EuroLeague, which instead takes more above-the-break threes (36% against 30%) and more short mid-range shots (21% against 16%). The rare EuroLeague corner three is worth more: 1.28 points per shot against 1.16 in the NBA. In both leagues threes are taken from just behind the league's own line, and beyond 2.5 m accuracy is the same; EuroLeague accuracy falls off faster on the mid-range wings and far beyond the arc (Figure 4).

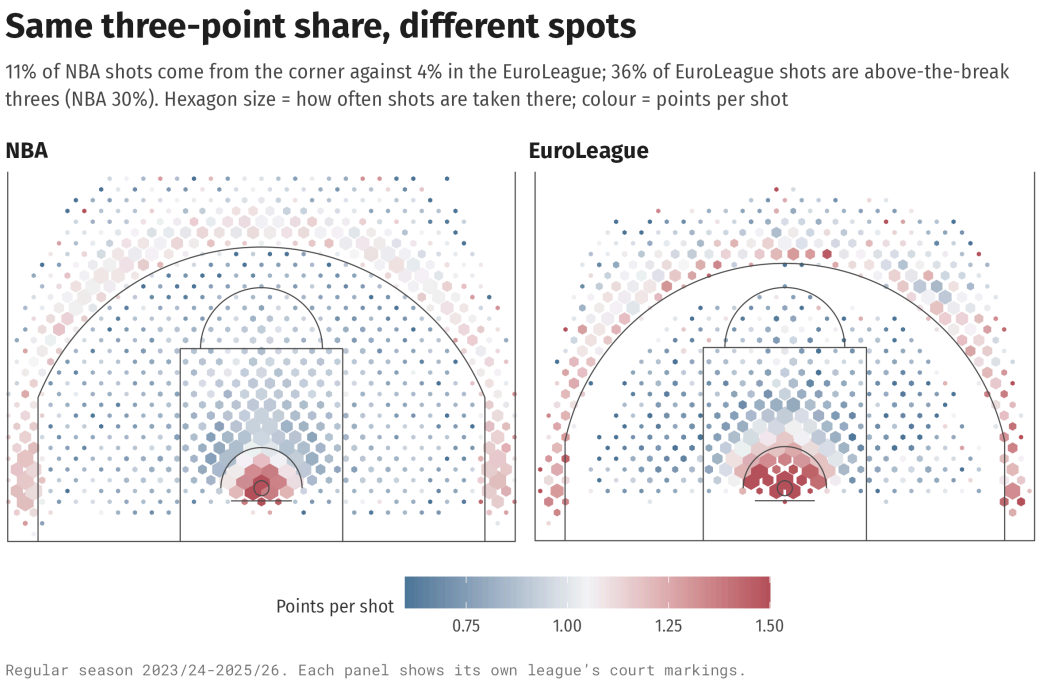


Figure 2: Shot charts, regular season 2023/24–2025/26. Hexagon size = how often shots are taken from there; colour = points per shot.

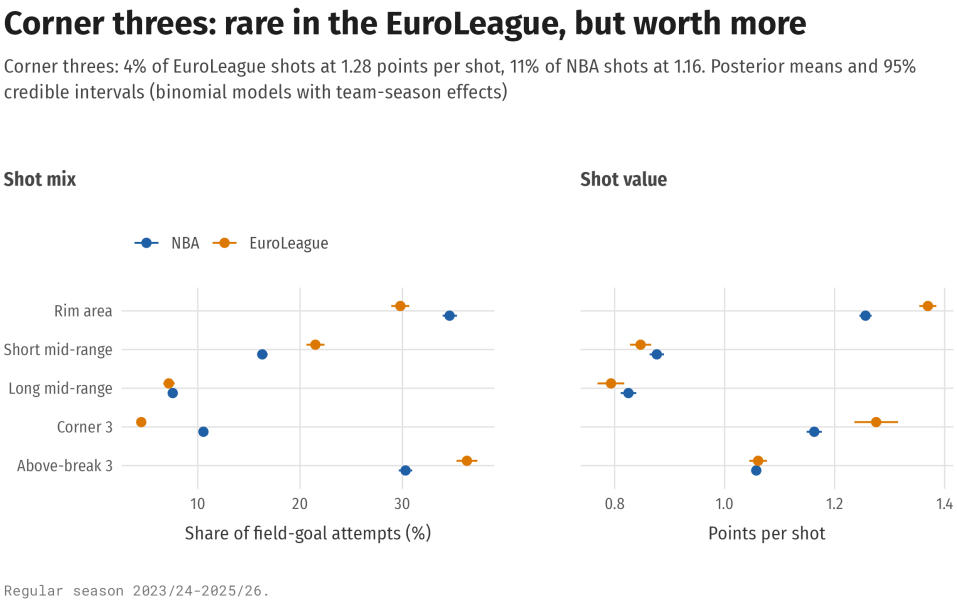


Figure 3: Shot share and value by zone for a typical team. Posterior means and 95% credible intervals from binomial models with team-season effects.

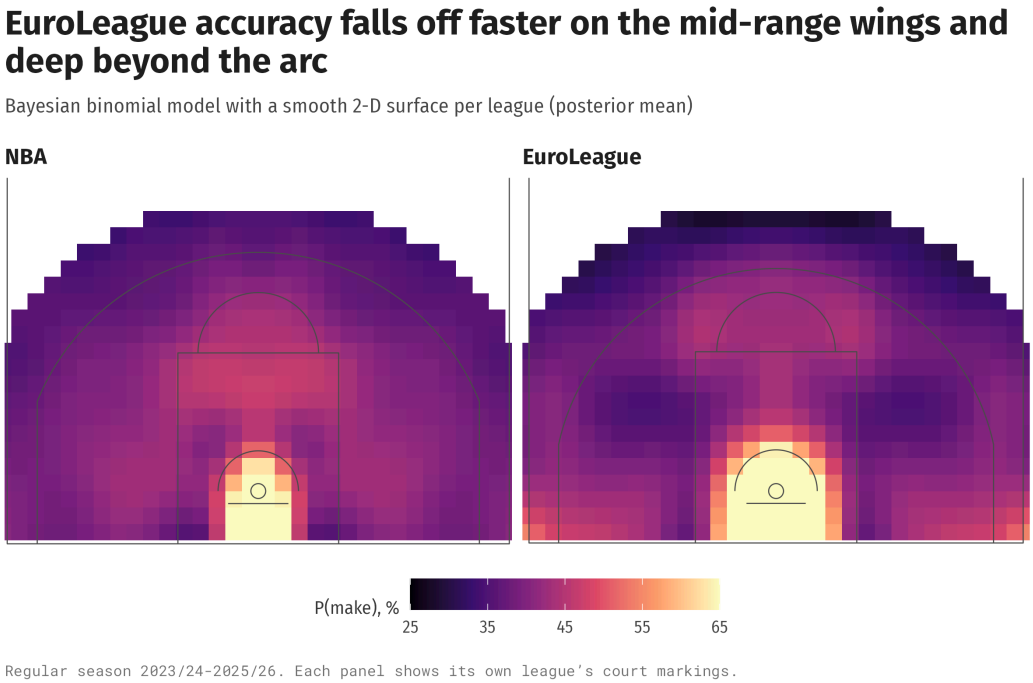


Figure 4: Modelled accuracy across the court: binomial model with a smooth two-dimensional surface per league (posterior mean).

4.2 How the leagues changed, and who follows whom (RQ2)

Replication. Fitted to the 2019 paper's own data, our implementation reproduces all 68 published estimates; the largest absolute difference is 0.013 (Figure 5).

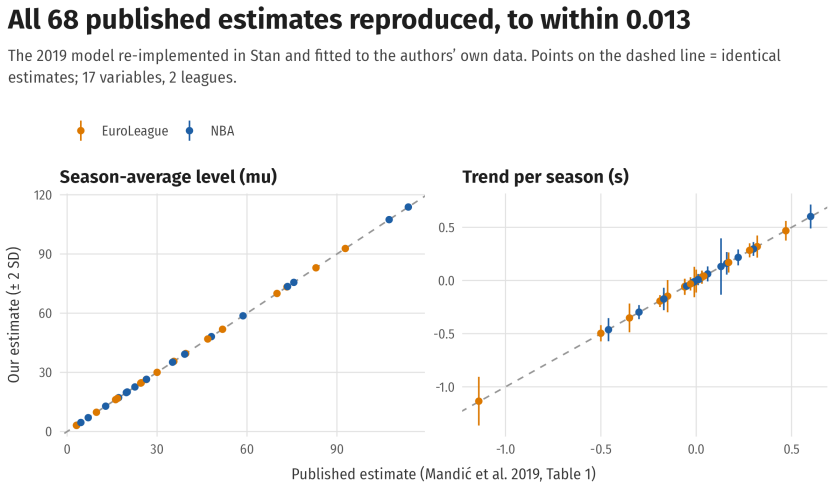
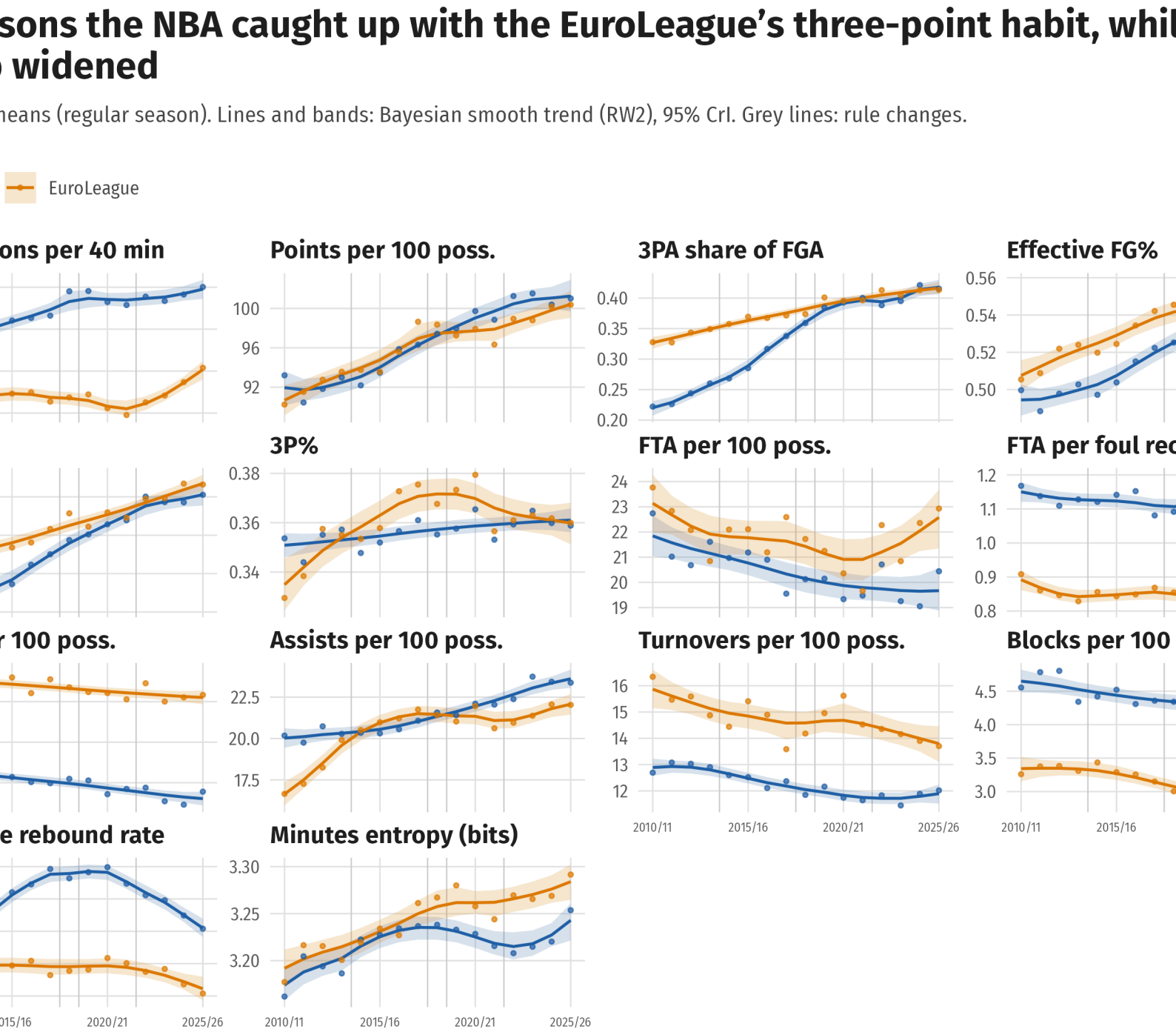


Figure 5: The 2019 model re-implemented in Stan and fitted to the authors' published data, against their Table 1.

Trends. Since 2010/11 both leagues have moved in the same direction on most metrics (Figure 6). The NBA closed an 11-point gap in three-point share: the gap went from −10.7 to 0.1 percentage points (probability that it shrank 1.00). Pace went the other way: the gap grew from 6.7 to 9.6 possessions per 40 minutes (probability that it shrank 0.003). Free-throw rate and offensive rebounding also drifted apart (Figure 7).



4/15 14-s reset after offensive rebounds (FIBA/EuroLeague); 2018/19 same rule in the NBA; 2019/20 COVID (NBA bubble, Eur
the regular season); 2022/23 NBA take-foul rule.

Figure 6: Season means (points) and Bayesian smooth trends (RW2, 95% credible bands), regular season 2010/11–2025/26. Grey vertical lines mark rule changes.

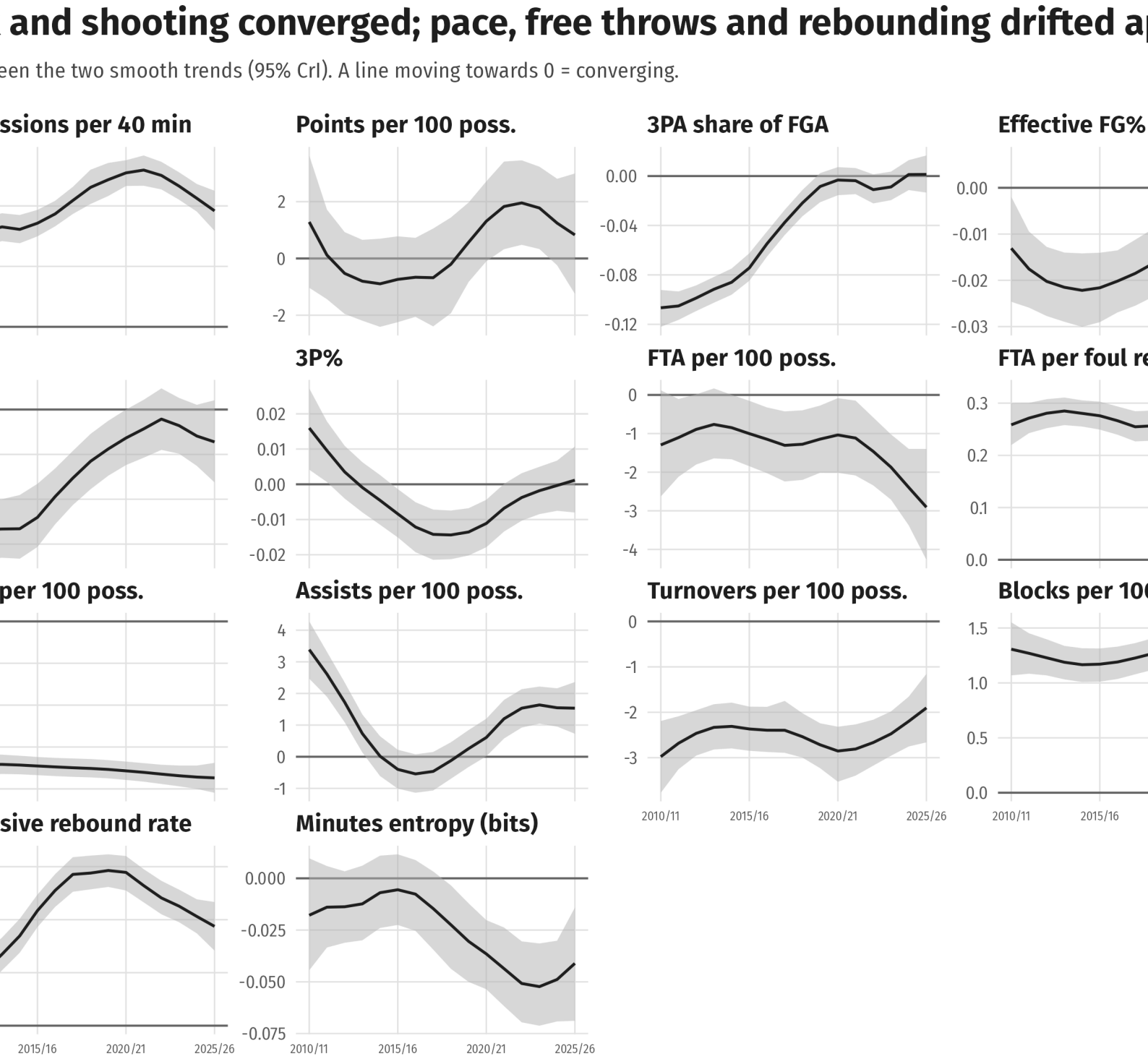


Figure 7: Gap between the leagues (NBA minus EuroLeague) over time, 95% credible bands. A line moving towards zero means the leagues are converging.

Lead–lag. For most metrics the shift model cannot say which league leads (Figure 8). The clearest signals concern three-point shooting: on three-point attempts per 100 possessions the leagues move in step (most probable lag 0, probability 0.50), and on the share of shots from three the EuroLeague is in step or one season behind (probability that it follows 0.65). On volume measures the EuroLeague of 2025/26 looks like the NBA of about 3 (points per 100 possessions), 4 (three-point attempts) and 5 (assists) seasons earlier (Figure 9).

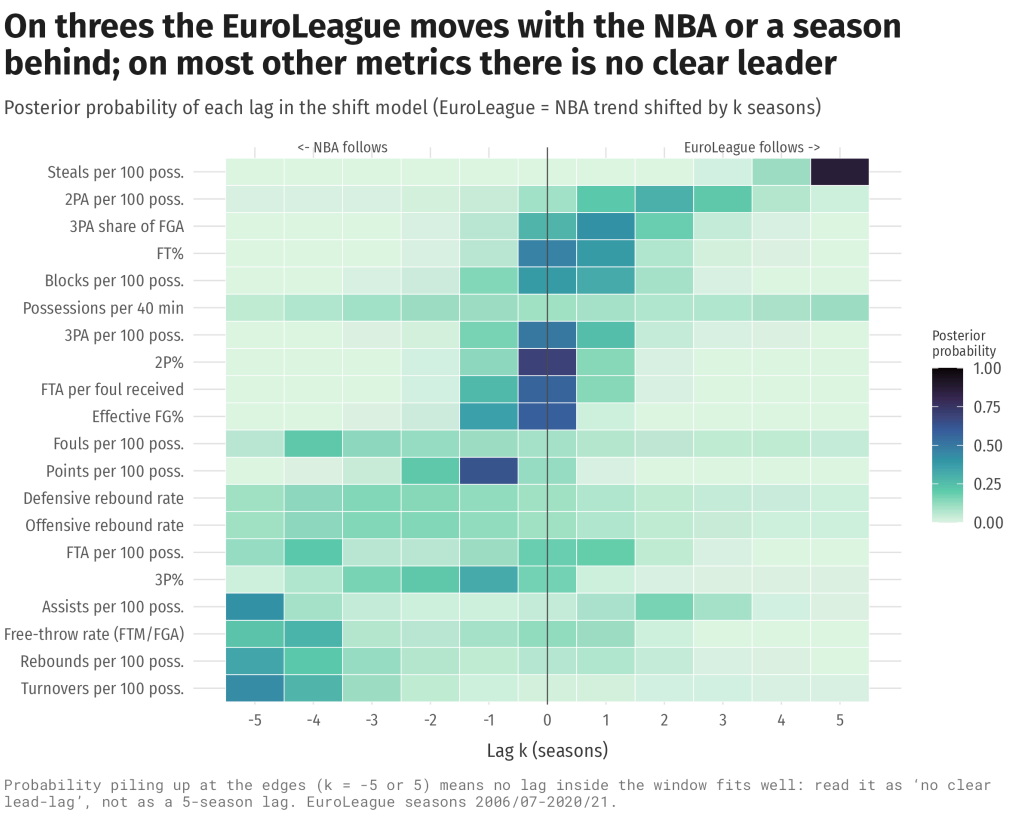


Figure 8: Shift model: posterior probability that the EuroLeague's trend equals the NBA's shifted by k seasons. Probability piling up at k = ±5 means that no lag inside the window fits well.

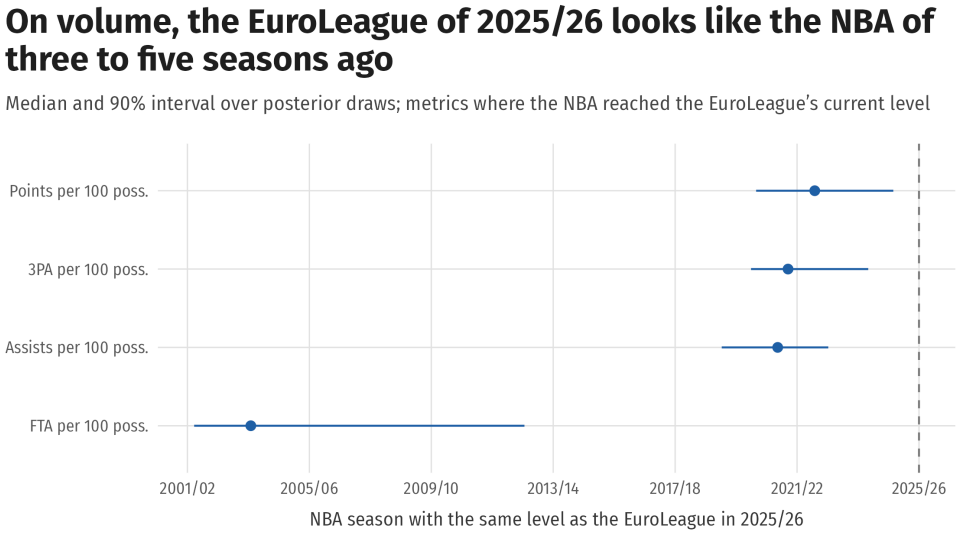


Figure 9: The NBA season in which the NBA first reached the EuroLeague's 2025/26 level (median and 90% interval over posterior draws), for metrics where it did.

4.3 The playoffs (RQ3)

In the playoffs both leagues slow down, foul more, take more free throws and shoot worse (Figure 10). The NBA's playoff shift points towards the EuroLeague's regular-season level on 10 of 21 metrics, mostly pace, fouls and free throws, and away from it on 8, including every field-goal percentage (Figure 11). The 2019 claim that NBA playoff basketball "looks European" therefore holds for tempo and physicality but not for field-

goal shooting. In shot selection, EuroLeague teams swap rim shots for above-the-break threes in the playoffs (−1.9 and +1.9 percentage points), while NBA teams move a little from the rim to the short mid-range.

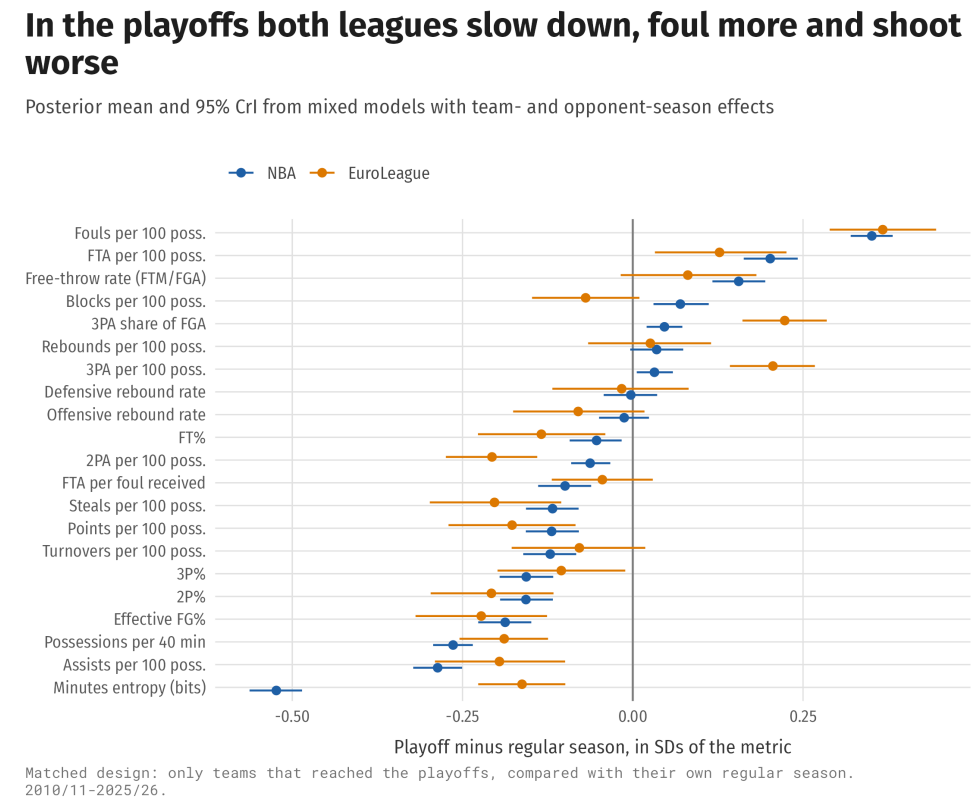


Figure 10: Playoff minus regular season for teams that reached the playoffs, standardised; posterior means and 95% credible intervals, 2010/11–2025/26.

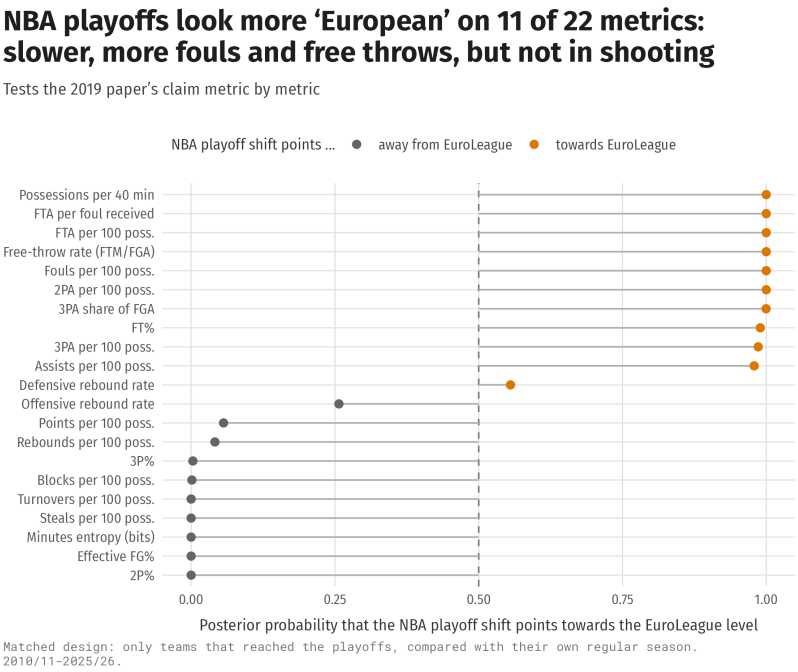


Figure 11: Does the NBA's playoff shift point towards the EuroLeague? Metric by metric.

4.4 Teams and players ahead of their time (RQ4)

The Houston Rockets were the NBA's most different team in seven straight seasons (2013/14–2019/20): almost no long two-pointers, many corner threes and free throws (Figure 13). Their 2012/13 and 2016/17 teams sit about four seasons ahead of the league, which later moved their way (Figure 12). In the EuroLeague, Real Madrid of 2013/14–2017/18 played the league's future: fast, assist-heavy, with few turnovers, almost no long two-pointers and many above-the-break threes. Teams behind their time include the 2022/23 Chicago Bulls, whose mid-range offence resembled an NBA team of about 2017/18, and Partizan 2022/23.

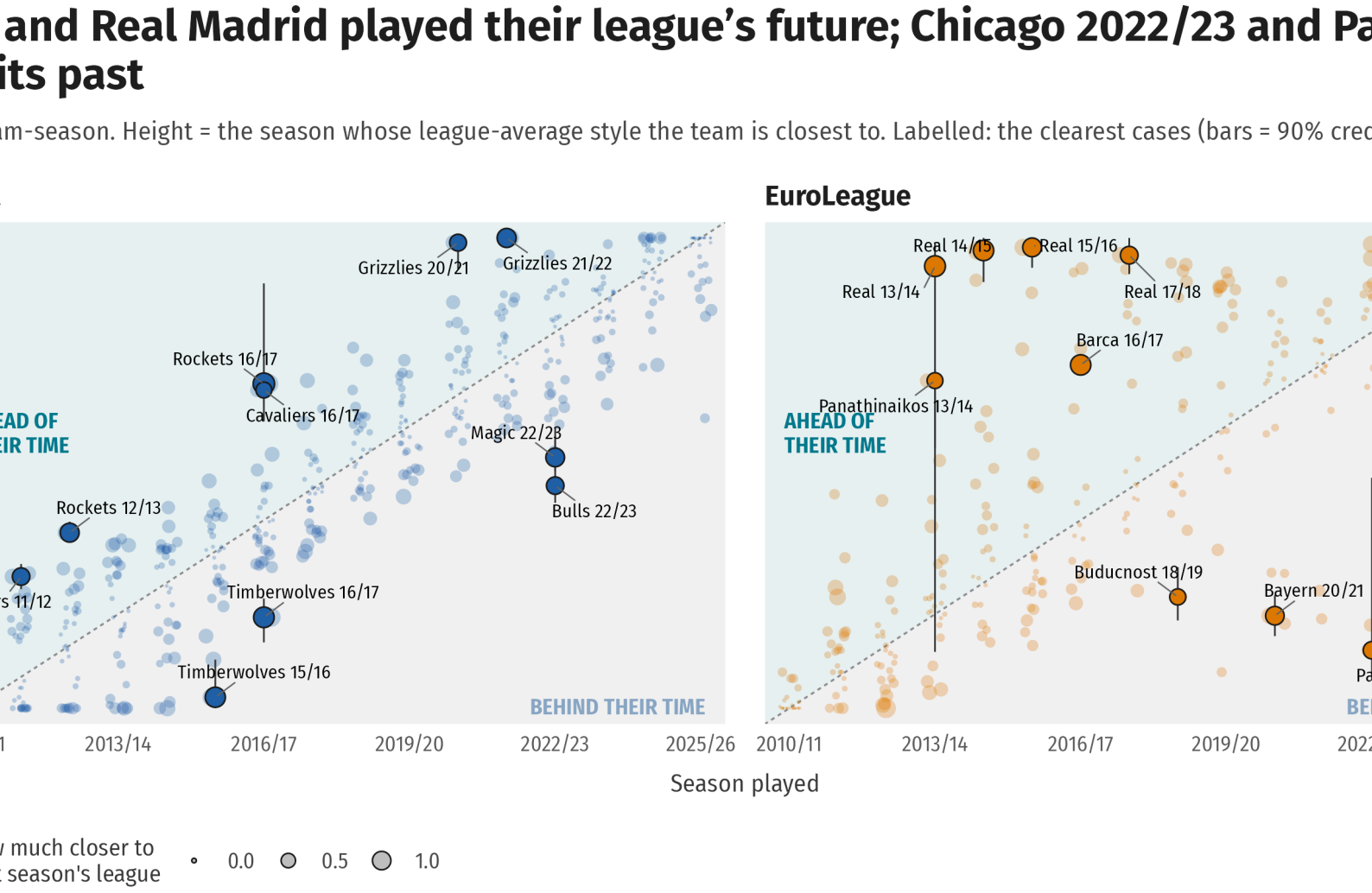


Figure 12: The time machine: the season whose league-average style each team-season is closest to. Above the diagonal = ahead of its time; labelled teams are the clearest cases (bars = 90% credible intervals).

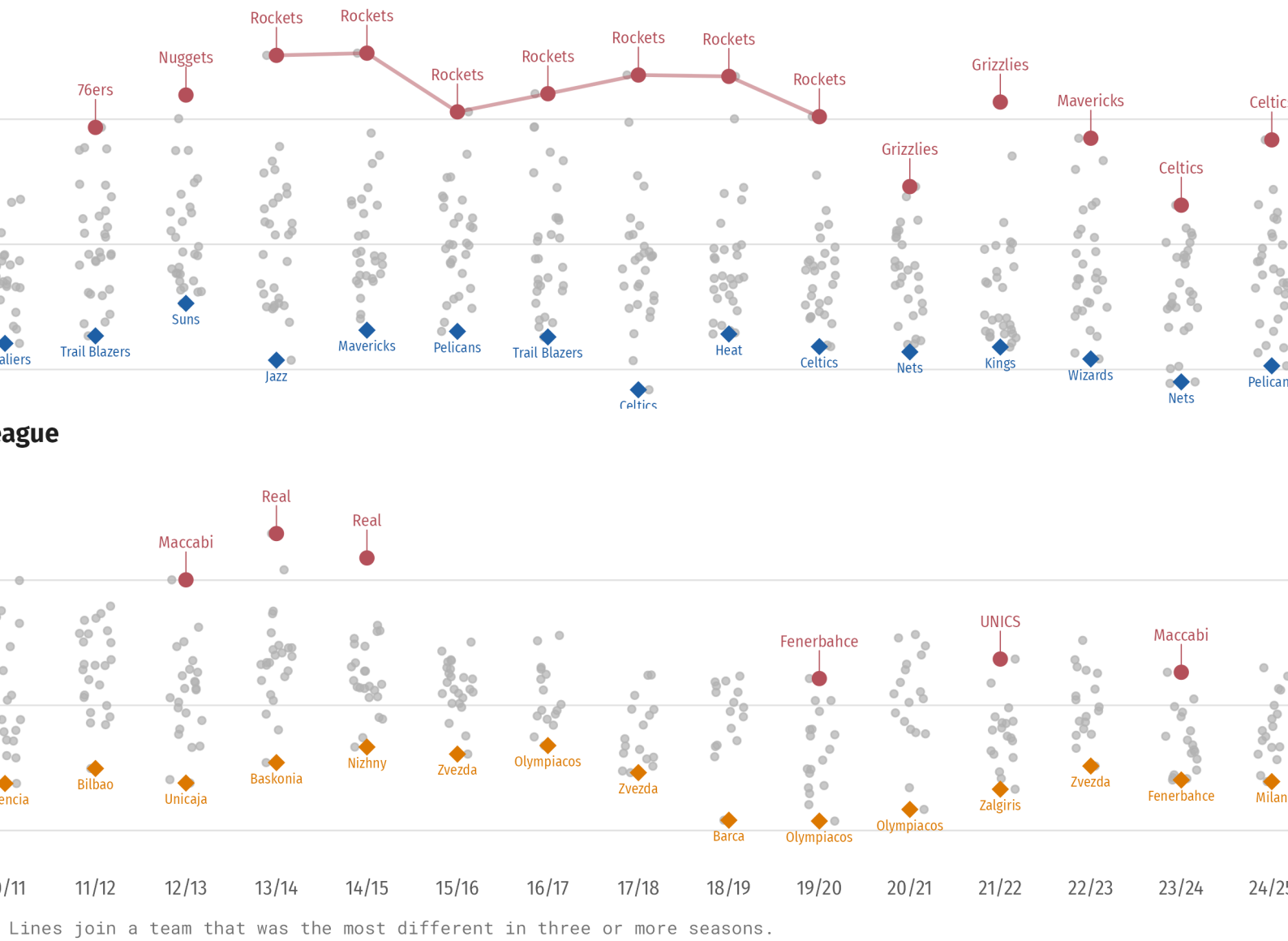
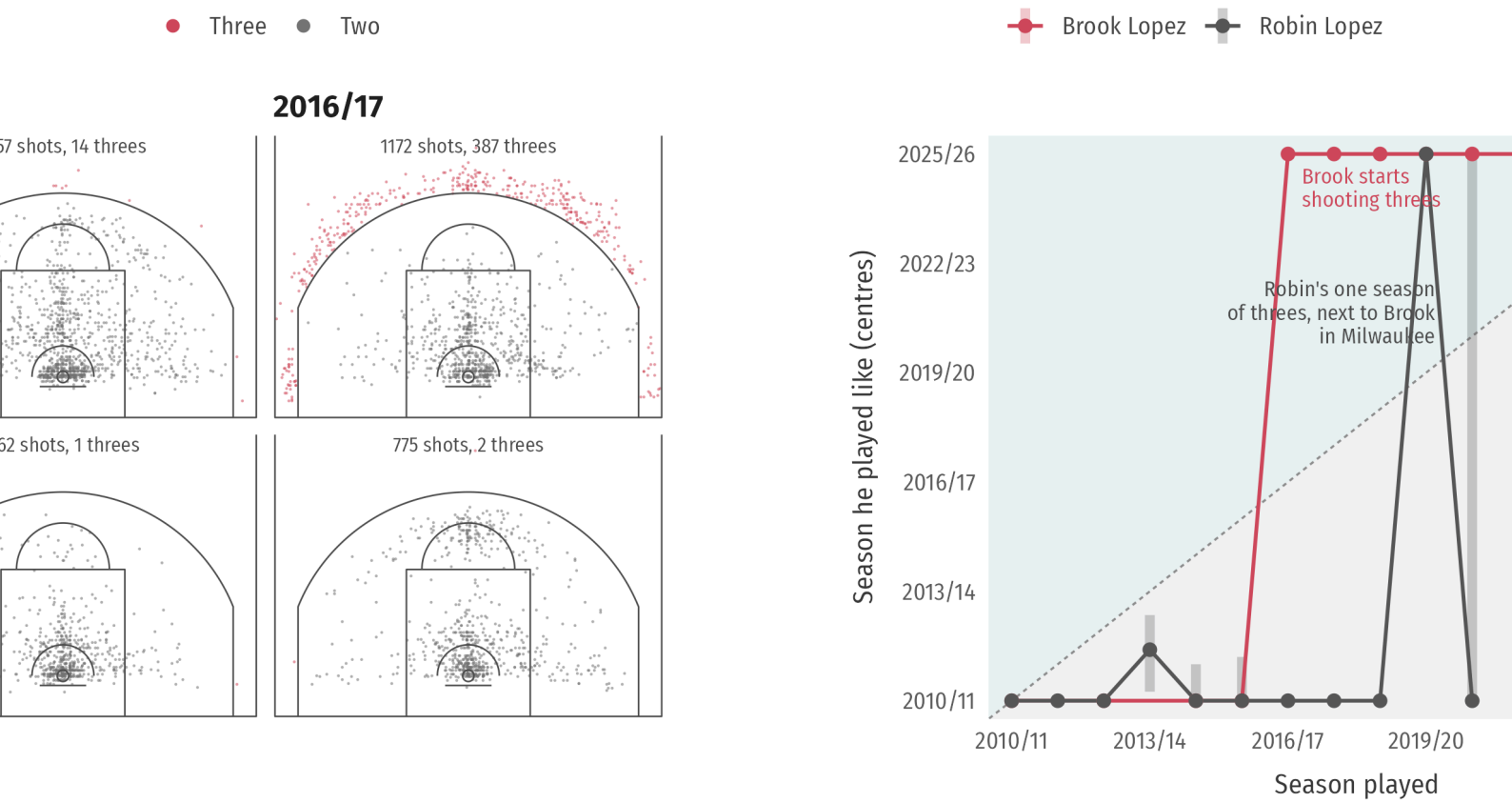


Figure 13: Each team's distance from its league's average style per season; red = the most different team of the season.

Among players, Nikola Kalinić was the EuroLeague's most typical player in six seasons. Brook Lopez is the clearest individual case of a player jumping ahead: from 14 three-point attempts in 2015/16 to 387 in 2016/17, moving from a 2010s centre to a centre of today in one summer, while his twin Robin stayed a 2010s centre (Figure 14). EuroLeague guards such as Juan Carlos Navarro, Miloš Teodosić, Sergio Llull and Vassilis Spanoulis sit ahead of their time for most of their careers.



es, different eras: Brook Lopez jumped a decade in one summer

opez moved out to the three-point line and jumped from the 2010s centre to the centre of today. Robin stayed near the rim for

ots. Right: time position among NBA centres (90% credible intervals).

Figure 14: Brook and Robin Lopez: shot charts before and after 2016/17, and their time positions among NBA centres (90% credible intervals).

5 Discussion

Sixteen seasons on, the picture of Mandić et al. (2019) needs two corrections. First, convergence is real but partial: the leagues now shoot alike and take the same share of their shots from three, but they have drifted apart in tempo and in how games are refereed. The NBA plays about ten more possessions per 40 minutes and the EuroLeague commits about seven more fouls per 100 possessions. Second, the idea that the NBA "became European" is better described as two leagues responding to the same pressures with different constraints. The EuroLeague took a larger share of its threes earlier, but the NBA takes more threes per possession, because it has more possessions and fewer turnovers. The equal three-point share is reached from different spots on the floor, shaped by the corner geometry of each court.

The lead–lag results describe timing, not influence. Smooth trends over about 25 seasons can often be shifted a few seasons either way and still fit, so for most metrics the honest answer is that the data cannot tell who leads. Where they can, the EuroLeague moves with the NBA or one season behind on three-point shooting, and resembles the NBA of three to five seasons ago on volume measures.

6 Limitations

Box scores and shot locations come from two scorekeeping systems: shots near the rim are recorded less precisely in the EuroLeague, and rebounds and turnovers are credited slightly differently. The EuroLeague's format changed several times (Top 16 until 2015/16, a single round robin since 2016/17, a play-in since 2025/26), and the 2019/20 season was cancelled after the regular season. Lead–lag results describe timing only; shared causes such as analytics, rule changes and player movement could produce the same pattern. The RQ4 results depend on which features define style; a different list could name different teams or players, and the posterior probabilities show where the choice is close. NBA possessions per 40 minutes before 2010/11 use regulation length, which only affects the lead–lag model.

7 Data and code availability

All code, Stan models, result tables and figures are at https://github.com/loukesio/nba-vs-euroleague (MIT licence). The data are downloaded from public sources by the scripts in `00_data/scripts/` and remain subject to their providers' terms. Figures were made with ggbasket (https://loukesio.github.io/ggbasket/).

References

Bürkner, P.-C. (2017). Brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Erčulj, F., & Štrumbelj, E. (2015). Basketball shot types and shot success in different levels of competitive basketball. *PLOS ONE*, *10*(6), e0128885. https://doi.org/10.1371/journal.pone.0128885

Gilani, S. (2021). *hoopR: Access men's basketball play by play data*. https://hoopr.sportsdataverse.org

Goldsberry, K. (2019). *SprawlBall: A visual tour of the new era of the NBA*. Houghton Mifflin Harcourt.

Kubatko, J., Oliver, D., Pelton, K., & Rosenbaum, D. T. (2007). A starting point for analyzing basketball statistics. *Journal of Quantitative Analysis in Sports*, *3*(3). https://doi.org/10.2202/1559-0410.1070

Mandić, R., Jakovljević, S., Erčulj, F., & Štrumbelj, E. (2019). Trends in NBA and Euroleague basketball: Analysis and comparison of statistical data from 2000 to 2017. *PLOS ONE*, *14*(10), e0223524. https://doi.org/10.1371/journal.pone.0223524

Oliver, D. (2004). *Basketball on paper: Rules and tools for performance analysis*. Potomac Books.

Rue, H., & Held, L. (2005). *Gaussian Markov random fields: Theory and applications*. CRC Press.

Theodosiou, L. (2026). *ggbasket: Learn from basketball games, NBA and EuroLeague*. https://loukesio.github.io/ggbasket/

Wood, S. N. (2017). *Generalized additive models: An introduction with R* (2nd ed.). CRC Press.